

Does an Empty Street Still Talk? A VR Study on Urban Liveliness and Perceived Safety

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Abstract

This study examines how perceived liveliness, implemented through street-level attributes and greenery, influences women's sense of safety in an urban street environment with a window facade, using virtual reality (VR). Fifteen women (aged 21-32) participated in a two-alternative forced choice (2AFC) task, semi-structured interviews, and questionnaires assessing perceived safety, presence, and simulator sickness. Six street conditions were combined into 15 paired comparisons, varying greenery density (none, low, high) and street-level attributes (present or not). Results showed that streets containing multiple attributes were perceived as safer than empty streets or streets containing only one attribute. This supports the hypothesis that cues of liveliness and potential human presence enhance safety perceptions. Additionally, street-level attributes had a stronger influence on perceived safety than greenery alone. Contrary to expectations, high-density greenery did not increase perceived safety compared to low-density greenery, possibly due to reduced visibility and feelings of entrapment. These findings highlight the context-dependent role of greenery and emphasise the importance of visibility in shaping perceived safety. Although limited by a small sample, the study demonstrates the usefulness of VR for investigating urban safety perceptions.

Keywords: *Virtual Reality, Perceived Safety, Urban Environment, Greenery Density, Women, Liveliness*

1. INTRODUCTION

Despite significant advances in modern urban development, a profound gender gap in perceived safety remains a defining and persistent characteristic of public spaces globally [4, 11, 15]. Recent data highlight that women consistently report drastically lower levels of safety than men, often feeling vulnerable in environments that traditional urban planning considers 'neutral' [6]. This safety gap is more than just a psychological issue; it directly affects how women use public space. Fear often forces women to change their daily routes or avoid going out at night, which limits their freedom and opportunities [3, 17, 23]. Therefore, research focusing specifically on these experiences is urgent. Virtual Reality (VR) offers a safe and controlled way to simulate urban environments and identify which design elements trigger feelings of danger or safety. In recent years, VR has been increasingly used in urban planning, as it enables the simulation of scenarios that would otherwise be difficult or unsafe to replicate at scale, making it a suitable medium for analysing participants' safety-related decisions [5, 13].

One of these otherwise difficult scenarios is

perceived safety. As mentioned in Gao et al. [7], perceived safety includes multiple different terms. Fear of crime, traffic safety, accidents and "an individual's general anxiety regarding external threats" are all part of the term [7]. For this project, this study will use the definition introduced by Mouratidis [18] and define perceived safety as "an individual's level of comfort and perception of risk within the environment" [18].

When discussing perceived safety, an important factor is perceived liveliness, which refers to indicators of human activity. This project is based on "Perceived Liveliness Impact on Sense of Safety: A Study on Women's Perceived Safety in Human-Lacking Urban Environments" by Ebba Rovig [22]. It investigates how qualities of liveliness, like street-level attributes, brick walls, windowed facades, and greenery, affect the perceived safety for women aged 22-31 in a VR environment. The present study uses the same Unity assets as Rovig's [22] study and builds upon its findings. Those findings reveal a hierarchy of street-level attributes affecting perceived safety more than windowed facades due to their greater impact on liveliness, and windows being more important than greenery [22].

Previous research has shown that more trees in an environment are associated with higher perceived safety [18] and that a greater canopy cover of trees is associated with lower rates of crime [8]. However, studies focusing on perceived safety in parks have shown that women tend to avoid zones with "extensive undergrowth and trees", as it creates a visual obstacle and conceals potential threats [16].

The present study aims to investigate how qualities of liveliness, specifically the presence of street-level attributes and the amount of greenery, affect the perceived safety of urban environments with windowed facades for women aged 21-32. The study aims to answer the following research questions:

RQ1: *How does perceived liveliness, experienced through VR, influence women's perceived safety in an urban environment with a*

window facade?

Hypothesis 1: The expected outcome based on Rovig [22] is that there is a connection between increased perception of liveliness and women's sense of safety, as it indicates human presence. Liveliness is shown through the presence of greenery and street-level attributes (cars, a bike lane, parked bicycles, a storefront, benches, and a bus stop).

RQ2: *How does the density of greenery and presence of street-level attributes impact women's perceived safety?*

Hypothesis 2: Based on Mouratidis [18] and Gilstad-Hayden [8], a higher density of greenery is expected to increase perceived safety compared to low or no greenery. Based on Rovig [22] and Newman [19], the presence of street-level attributes (e.g. cars and bicycles) is also expected to increase perceived safety. Overall, street-level attributes are hypothesised to have a stronger influence on perceived safety than greenery alone [22].

2. BACKGROUND

2.1 Perceived liveliness

Perceived liveliness, meaning indicators of human presence, is an important factor when examining perceived safety in an urban environment. Some of these attributes, which contribute to perceived liveliness, are bikes, storefronts, cars, bus stops and bike lanes, as shown by [22]. Other important factors illustrated in research include schools, parks, lighting and open spaces [20].

2.2 Greenery and its impact on perceived safety

Several studies have examined the relationship between greenery and perceived safety [10, 14, 21]. While greenery is generally associated with positive perceptions during daytime, the same environments are often avoided after dark. Dense vegetation close to walking paths can create feelings of entrapment, indicating that perceived safety depends not only on the

presence of greenery but also on its spatial arrangement [21]. Research suggests that more unified and orderly vegetation is perceived as safer than fragmented or randomly dispersed greenery [14, 10].

Research has also shown that perceived safety during the day is positively correlated with open views, more colours, and good maintenance of vegetation [26]. Regarding the maintenance of vegetation, well-trimmed greenery is perceived as safer than its wild-grown alternative [26]. This would be in line with the Broken-Windows theory, suggesting a relationship between environmental disorder and crime, where “one unrepaired broken window is a signal that no one cares, and so breaking more windows costs nothing” [24].

2.3 Prospect-Refuge Theory

Central to the field of urban and landscape design is the Prospect-Refuge theory, first introduced in 1975 by Appleton [2]. It is defined as “an evolutionary preference for environments that offer safety (refuge) while also having the ability to survey your surroundings for resources, potential threats, and other organisms” [1]. Therefore, the theory highlights a conflict between the need for hiding spots and the need for open environments that can easily be scanned for potential threats.

3. METHOD

3.1 Participants

15 women aged 21-32 were recruited through a convenience sample. All participants were students at KTH Royal Institute of Technology.

3.2 Test Environment

The study was conducted in an immersive VR environment in Unity (Version 2021.3.9f1), with the Meta Quest 2 Headset as the device. The model file used was based on Rovig [22] and modified to include different greenery densities. The VR scene depicts an urban 90-degree street corner, where two streets extend either to the left or the right from the partici-

pant’s standing point. See Figure 1.



Figure 1: Participants’ starting point shown from a high angle.

The two different attributes create six possible streets, where each street could either be shown on the left side or the right side (see Figure 2). The possible streets can be seen in Table 1.

Table 1: Illustrates the possible streets

Possible streets
Empty
Street-level attributes (SLA)
Street-level attributes (SLA) + low-density greenery (LDG)
Street-level attributes (SLA) + high-density greenery (HDG)
Low-density greenery (LDG)
High-density greenery (HDG)

This is a within-subjects design. The factorial structure is: 3 (Greenery: none vs low-density vs high-density) $\times$ 2 (Street-level attributes: present vs absent). Participants are exposed to all combinations, resulting in a 3 $\times$ 2 repeated-measures design. These streets are paired as corners, so there are 15 combinations where participants are asked to choose between two streets (see Figure 3).

3.3 Pilot Study

Two rounds of pilot studies were conducted. In the first pilot test, two members of the research group acted as participants. The purpose was



Figure 2: Example of a street combination from a high angle.

to verify whether the entire technical architecture functioned properly, including functions such as tracking the VR headset, the display of the Unity scene, and the timing of the beep sound alerts. The second pilot test was run on one external participant using the same inclusion criteria as in the main study. Beyond testing functionalities, this round focused on evaluating the recruitment process, the usability of follow-up interviews, and questionnaires. However, due to difficulties with the VR headset, this was tested with a screen only and not in VR.

3.4 Procedure

Before starting the test, each participant signed an informed consent form regarding what data would be collected and stored, possible risks of motion sickness, and how to withdraw consent.

At the start of the test, participants were asked to stand in the middle of a physical room with their standing point corresponding to a fixed starting position within the virtual environment. See experimental setup in Appendix A. Before putting on the VR headset, participants were all given the same scenario: “You are walking home by yourself. You will be standing at a corner with two streets in front of you, both can take you home. When you hear the beep, pick the street that you perceive as safer”. It was a two-alternative forced choice (2AFC) method. They could pick the street by saying “Left” or “Right”, or by pointing to-

ward the street that made them feel the safest. Participants were encouraged to trust their gut feelings when deciding. After 12 seconds, a beep indicated that a decision had to be made.

15 combinations were created using this equation:

$$\binom{6}{2} = \frac{6 \times 5}{2}$$

The participants viewed 15 different settings, covering every possible street combination without repeating. There was a delay of a few seconds between viewing each combination. To prevent left-right bias or learning effect, the order of these pairs, as well as the positioning of left vs. right, was randomly selected for each person. Since 15 is an odd number, an exact 50/50 split was impossible for a single participant. Therefore, the balance was alternated: one participant received an 8 vs 7 split, and the next received a 7 vs 8 split. This method ensured that, across the entire sample, the spatial distribution was perfectly even.

The interview questions and questionnaires were done after the VR testing. When completing all tasks, each participant was asked to remove the VR headsets, take a seat, and complete a semi-structured interview with one group member. The interview questions focused on how they perceived the attributes in terms of perceived safety. The interview was recorded with the participant’s consent. After the interview, the questionnaires were done in Google Forms as a computer-based survey. Three questionnaires were completed: 1) perceived safety, 2) presence, and 3) simulated sickness. To quantify perceived safety, a questionnaire was created that listed items that asked how safe participants felt under certain conditions. The conditions were concerning each attribute (street-level attributes, low-density greenery, high-density greenery) and a street without any attributes (only the window facade). This was ranked on a 7-point Likert scale, 1 being “not safe at all”, and 7 being “very safe”. See Appendix C. To measure how present the participant felt and therefore how applicable the results would be to real life, the

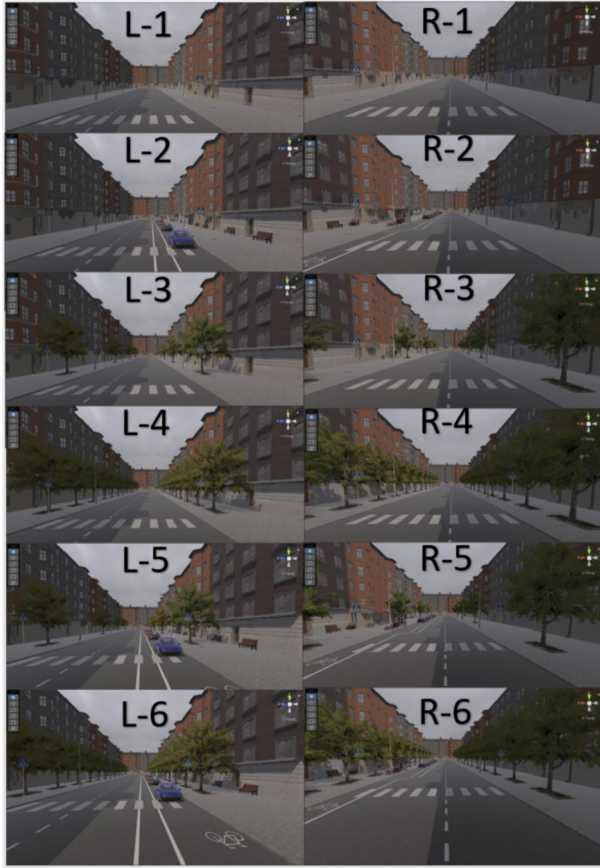


Figure 3: Six possible streets, L: streets on the left; R: streets on the right.

Presence Questionnaire by Witmer et al. [25] was used. To measure motion sickness, the Simulator Sickness Questionnaire (SSQ) was used [12].

3.5 Delimitations

The following has been excluded from this study:

- Comparison between window facades and solid walls.
- Perceived safety under nighttime and varied lighting conditions.
- Real-world and simulated social presence, including human behaviour and virtual agents.
- Impact of auditory cues, including the absence of ambient urban sound.
- Cultural, geographic, and socioeconomic influences on safety perception.

3.6 Data Analysis

During the test, both qualitative and quantitative data were collected. Qualitative data were collected through the interviews and analysed using thematic analysis. Quantitative data were collected through the 2AFC and questionnaires. The two alternative forced-choice (2AFC) data were analysed using a binomial test, which is a statistical approach for analysing 2AFC and paired-comparison tasks, where each trial results in a binary outcome and a change in performance corresponds to a proportion of 0.5 [9]. The binomial test was used to test whether participants preferred one street over the other, rather than choosing randomly. A one-sided (greater than) alternative hypothesis was used, as the hypothesis was directional, predicting that increased liveliness, implemented through attributes such as high-density greenery and street-level attributes, would increase perceived safety and thus be chosen more frequently than less lively street environments. To measure if handedness had created a left-right bias, contingency tables in the form of a chi-squared test, Fisher's exact test and an odds ratio confidence interval were done. The analyses were done in Jasp. For the questionnaires, descriptive statistics were used, using Google Sheets, where the mean and SD were analysed.

4. RESULTS

4.1 Pilot Study

Completing the pilot study provided insights that made the actual tests more streamlined. The aspects that were changed:

- Changed the number of street combinations from 12 to 15, as 12 did not incorporate all possible combinations.
- Optimised the Unity file so it was faster to combine streets while doing the test.
- Reduced participants' decision time from 30 seconds to 12 seconds, as feedback showed that 30 seconds was too long since participants thought about their de-

cisions for a long time before picking, which does not reflect reality.

The data from the first pilot study was not analysed, as it was collected on the members of the group, creating bias. The data from the second pilot study was also not analysed, as it was not done using VR.

4.2 Test

The 2AFC data showed a strong preference toward streets with several attributes (both street-level and greenery) and a preference toward street-level attributes over greenery relating to an increased perceived safety. See Figure 4.

The left-right bias was tested to check if the randomisation worked. The testing showed that left- and right-handed participants chose the left and right options at roughly equal rates, suggesting that handedness did not bias their decisions. A chi-square test showed no significant association between handedness and participants' left/right decision, $\chi^2(1) = 0.51$, $p = .48$. Fisher's exact test similarly indicated no significant relationship between handedness and the left or right decision ($p = .60$). The odds ratio's confidence interval included 1, indicating that handedness did not predict which option participants selected. See Tables in Appendix D.

4.3 Interviews

After being exposed to the stimuli in VR, interviews were conducted with all participants. Using thematic analysis, their answers were coded and grouped into several themes. These themes helped explain why participants chose one street over another. The findings mainly relate to four areas:

- Street-level attributes (urban infrastructure)
- Attitudes toward the density of greenery
- Visibility and openness of the space
- Others (neighbours, daytime)

Below each theme is a description based on the

coded data.

93.3 % of the participants stated that streets with general or certain attributes, such as cars, bikes, shops, benches, or a bus stop, felt safer than an empty street (streets without attributes). Even if there were no real people in the VR scene, these elements suggested that "people usually come here". They felt more comfortable when the street looked used and maintained. Overall, participants believed that a lively street meant fewer risks, while an empty street felt unpredictable and unsafe.

Greenery played an important but complex role in safety. Most participants liked low-density greenery and said it made the street feel friendly and pleasant. However, high-density greenery sometimes made participants feel unsafe. When trees blocked the view or created dark or covered areas, participants worried that someone could hide behind them. Across interviews, the preferred order of greenery density for safety, from most favoured, was:

low-density greenery → high-density greenery → no greenery (and no street-level attributes)

This shows that greenery is not always positive; visibility still matters. Furthermore, some participants stated that greenery made them feel safer, but most were not sure why. Some stated that it provided an emotional safety as participants felt that well-kept greenery shows that people care about the neighbourhood.

Whether participants can clearly see their surroundings is also a crucial factor for their perceived safety. Participants felt safer on wider streets and streets with sidewalks and a clear view. One particular consideration involved the orientation of the vehicle. Some participants noted that a vehicle facing them would cause unease. However, they believed this unease would diminish if the street was sufficiently wide, as they would have ample time to react to unexpected events. In contrast, narrow or visually blocked streets created uncertainty. This shows that participants were not only looking at the presence of attributes but

also interpreting their possible motion or implied direction. This adds an additional layer to how street elements influence safety; participants reacted to dynamic cues even in a static VR environment. Furthermore, all participants indicated that their decisions in the VR environment would correspond to the choices they expected to make in a real-world environment.

Some of the participants also mentioned social and time factors. For instance, they would feel generally safer when in a safe community or during the daytime. These cues were not directly part of the VR environment and our experiment, but participants naturally used them to interpret the street's safety. For example, the neighbourhood looked residential and quiet, which made it seem safer.

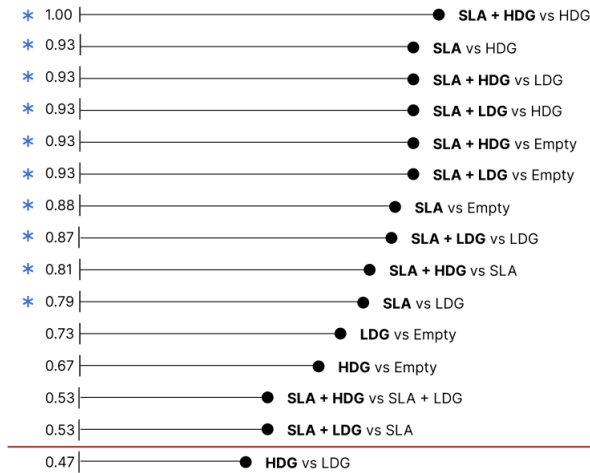


Figure 4: Lollipop chart of each street combination compared against each other. The bolded street signifies the street that was predicted to be chosen, according to the hypotheses. The y-axis shows the percentage that the bolded street was chosen. The blue * signifies a significant result of $p < 0.05$. All results aligned with the hypotheses except for the last street combination, where LDG was preferred over HDG; however, this result was not significant.

4.4 Questionnaires

The participants reported a medium to high perceived safety ($M = 4.25$, $SD = 1.71$). Interestingly, the streets with street attributes were

perceived as the safest ($M = 5.73$, $SD = 1.24$) while an empty street was perceived as the least safe ($M = 2.33$, $SD = 1.01$). All participants reported a moderate presence ($M = 4.92$, $SD = 1.47$). The participants reported almost no simulator sickness ($M = 0.28$, $SD = 0.57$). See Table 2.

Table 2: Descriptive Statistics

	<i>M</i>	<i>SD</i>
Presence	4.92	1.47
Simulator Sickness	0.28	0.57
Perceived Safety	4.25	1.71
SLA	5.73	1.24
LDG	4.73	1.00
HDG	4.20	1.42
Empty	2.33	1.01

5. DISCUSSION

The results from the two-alternative forced-choice (2AFC) task, the interviews, and the presence questionnaire consistently showed that participants perceived streets containing several attributes as safer compared to empty streets. However, the findings also indicate that not all attributes contribute equally to perceived safety; rather, a hierarchy of perceived safety emerges. This directly addresses RQ1, which asks how perceived liveliness influences women's sense of safety in urban environments. Across the 2AFC, interviews, and perceived safety questionnaire, street-level attributes were shown to have a greater impact on perceived safety than greenery alone. In the interviews, several participants stated that street-level attributes contributed more strongly to a sense of liveliness and to the perception that other people might be present in the environment, despite the absence of actual human figures. These findings support Hypothesis 1, which predicted that increased perception of liveliness, operationalised through greenery and street-level attributes, would positively influence perceived safety by acting as an indicator of human presence. The finding also aligns with Rovig [22], who similarly reported that street-level attributes exert

the strongest influence on perceived safety.

Contrary to the hypothesis, high-density greenery did not increase perceived safety more than low-density greenery, as indicated by the 2AFC, interviews, and the perceived safety questionnaire. In fact, in the 2AFC participants selected low-density greenery as safer in 53 % of the time. This result partially contradicts Hypothesis 2, which predicted that higher density greenery would lead to increased perceived safety compared to low-density or an absence of greenery. One possible explanation for this is that the sidewalk was partially obstructed by high-density greenery, whereas it remained unobstructed in the low-density condition, which several participants identified as important for their sense of safety. Furthermore, when high-density greenery was combined with street-level attributes, some of these attributes, such as storefronts, were partially occluded by trees. This reduced visibility may have diminished the contribution of street-level attributes to liveliness and safety, thereby counteracting the expected positive effect of increased greenery.

Another possible explanation for high-density greenery not being perceived as safer relates to the prospect-refuge theory. High-density greenery may provide refuge in the sense that it offers places to hide, but it simultaneously reduces prospect, or the ability to visually survey the environment. While the presence of refuge can increase comfort in some contexts, the reduced visibility may also increase uncertainty, as others could remain concealed within the vegetation. Low-density greenery does not involve the same trade-off, as it allows both visual openness (prospect) and limited opportunities for refuge. Additionally, as noted by Rahm et al. [21], vegetation placed close to sidewalks can contribute to feelings of entrapment, which may further reduce perceived safety in environments with dense greenery. These findings suggest that greenery contributes to perceived safety only under conditions where it does not compromise visibility or spatial openness, highlight-

ing the context-dependent role of greenery in supporting liveliness in urban environments.

It should be noted, however, that the preference for low-density greenery over high-density greenery was not statistically significant, suggesting that greenery density alone may not be a strong determinant of perceived safety. This nuance is particularly relevant for RQ2, as it indicates that greenery density does not independently drive perceived safety in the absence of other attributes. This is particularly noteworthy given that streets combining high-density greenery with street-level attributes were perceived as slightly safer than those combining low-density greenery with street-level attributes. However, this difference was not statistically significant, and the preference for high-density greenery in this comparison was modest, at only 53 %. This suggests that any additional safety benefits of high-density greenery may be weak or highly context-dependent, especially when street-level attributes are already present.

Across all measures, empty streets were consistently perceived as the least safe option when compared to streets containing any form of attributes. This finding supports the hypothesis that both greenery and street-level attributes contribute to perceived liveliness, which in turn increases perceived safety. In relation to RQ1, this underscored the importance of liveliness as a central mechanism through which environmental features influence safety perceptions. Nevertheless, some inconsistencies were observed between the 2AFC task and the interview data. During interviews, several participants stated that they never chose the empty street, as they consistently perceived streets with more attributes as safer. However, the 2AFC data revealed that the same participants occasionally selected the empty street over streets with attributes. This discrepancy may indicate momentary uncertainty during the forced-choice task, accidental selections, or post-hoc rationalisations during the interviews.

Based on the combined findings, the hierarchy

of perceived safety from highest to lowest can be visualised in Table 3.

Table 3: Illustrates the hierarchy of attributes for perceived safety, from highest to lowest.

Hierarchy of attributes for perceived safety
Street-level attributes + high-density greenery
Street-level attributes + low-density greenery
Street-level attributes
Low-density greenery
High-density greenery
Empty

Overall, the research questions were successfully addressed, demonstrating that perceived liveliness plays a central role in shaping women’s safety perceptions in urban environments. While street-level attributes reliably enhance perceived safety by signalling human presence and activity, greenery alone had a weaker and more ambiguous effect. Furthermore, handedness did not appear to influence participants’ left-right decisions in the 2AFC task. The results showed no systematic preference for selecting the left or right option, correlating with the participants’ dominant hand. This suggests that the observed decisions were driven by attributes in the street environments. Although the findings are not highly generalisable due to the small sample size ($N = 15$), the study confirms insights stated in Rovig [22] regarding the influence of street-level attributes and greenery on women’s perceived safety in urban environments. The results suggest that street-level attribute is a strong cue for liveliness and may be more influential when paired with greenery. This has potential implications for urban design, indicating that the visibility and presence of attributes may matter more than their quantity.

To confirm the reliability of the results, it is important to assess the participants’ experience in VR. The questionnaire data support the validity of the simulation as a tool for measuring

perceived safety. First, participants reported a moderate-to-high sense of presence ($M = 4.92$, $SD = 1.47$). This indicates they felt immersed in the environment. Importantly, most participants felt that their decisions in VR would match their choices in real life. This suggests that the results are transferable and likely reflect the actual navigation strategies women use in urban spaces. An important strength of using VR is the ability to easily and precisely manipulate levels of greenery, which would be difficult or impractical to achieve in real-world environments. Second, simulator sickness was negligible ($M = 0.28$, $SD = 0.57$). This low score confirms that physical discomfort did not affect the participants’ judgment, ensuring that their choices were driven solely by the environmental attributes rather than nausea or dizziness.

5.1 Project Assessment

Overall, the project met its main goals of exploring how liveliness and greenery density influence perceived safety in a controlled VR environment. The timeline was followed well, and the group was able to complete all required stages, including VR development, pilot testing, participant recruitment, data collection, and analysis. The 2AFC method worked as intended, and the interviews provided rich explanations that supported the behavioural results. The randomisation procedure for left/right choices also worked well, reducing potential bias. However, some risks and issues appeared during the project. The main challenge was the VR headset, which sometimes had tracking instability or connection interruptions, especially during the early pilot phase. Additionally, some Unity elements, such as car orientation and the distance to specific objects, could not be fully controlled and may have influenced participants’ decisions unintentionally. For example, one participant reported that car placement affected her sense of safety; the car facing towards her seemed less safe than the one facing away from her. Despite these issues, the team managed to adapt quickly.

5.2 Limitations

A key limitation of this study is the small sample size ($N = 15$), which limits statistical reliability, representativeness, and the ability to draw statistically significant conclusions. Although the sample was sufficient to identify general patterns, larger and more diverse samples are needed to improve generalisability, particularly across the full 21-32 age range. Furthermore, the participant pool consisted exclusively of KTH students, resulting in a relatively homogeneous group in terms of age, educational background, and familiarity with technology, despite individual differences in cultural background. Another limitation was the uniformity of the street design in Unity. Due to the structure of the street layout, the street-level attributes of the two streets cannot be entirely consistent. For example, the orientation of vehicles cannot be fully controlled as a variable. Additionally, the distance from participants to the bus stop will influence their perceived safety, which was not consistent in both the left and right environments. Furthermore, participants' movement was restricted, as they could not walk too far from the starting point. Some attributes might be too far to be seen or covered by the greenery. The VR environment did not include natural sound (city noise), interaction, nighttime or human agents. These missing aspects would have made the testing a more immersive experience for participants, and perhaps closer to a real-world environment. The perceived safety questionnaire was not formally validated, and some items could be ambiguous to participants. Combined with the small sample, this weakened the robustness of the quantitative results. Another limitation is post hoc rationalisations, as it's possible that participants made up reasons for their decision after the fact and that the reasons they gave did not reflect reality. They were also encouraged to go with their gut feeling while making the 2AFC decisions, as that reflects how decisions are made in real life. This could also have contributed, as they were told not to think about their decision-making in too much detail dur-

ing testing, but were asked to explain their reasoning afterwards.

5.3 Future Research

Future research could improve this study in several ways. A larger participant pool would increase the reliability of the results and make it possible to compare safety perceptions across different groups of people. The amount of greenery could be explored in more detail, especially a moderate or middle level of greenery. Perceived safety could also be compared under different lighting conditions, such as darkness or nighttime environments. Since lighting strongly influences how safe a street feels [21], testing the same scenes in low-light conditions could provide more realistic insights. Other environmental cues could be studied more closely, including car direction. The orientation of vehicles appeared to affect participants' comfort, and considering this variable separately could help clarify its role in perceived safety. In addition, adding virtual agents, such as pedestrians, would allow researchers to examine how social presence influences safety judgments. Introducing movement and sound could further increase realism. Moving cars, walking agents, or ambient sounds such as traffic noise and footsteps may trigger stronger emotional responses and lead to different safety evaluations. Including these elements could help future studies better reflect real urban environments.

6. CONCLUSION

This study investigated how perceived liveliness, operationalised through street-level attributes and varying densities of greenery, influences women's perceived safety in an urban street environment with a window facade using virtual reality. Across the 2AFC, qualitative interviews, and questionnaires, the findings consistently indicate that environments containing multiple cues of liveliness are perceived as safer than empty streets or streets with a single attribute. In particular, street-level attributes such as cars, cafes, and bus

stops had a stronger and more reliable effect on perceived safety than greenery alone, supporting the notion that cues implying human presence play a central role in shaping safety perceptions.

Contrary to expectations, high-density greenery did not enhance perceived safety compared to low-density greenery. Instead, dense vegetation was sometimes associated with reduced visibility and feelings of entrapment, highlighting the importance of visual openness and prospect in urban design. These results emphasise that the contribution of greenery to perceived safety is highly context-dependent and that its benefits may diminish when it obstructs sightlines or obscures other indicators of liveliness. While the small and relatively homogeneous sample limits the generalisability of the findings, this study demonstrates the value of VR as a methodological tool for examining safety perceptions in controlled yet immersive settings. Overall, the findings suggest that urban design strategies aiming to improve women's perceived safety should prioritise visible signs of human activity and carefully balance greenery with openness and visibility.

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APPENDIX

A. TEST ENVIRONMENT



Figure A.1: Experimental setup

B. STREET COMBINATIONS



Figure B.1: HDG vs HDG + SLA

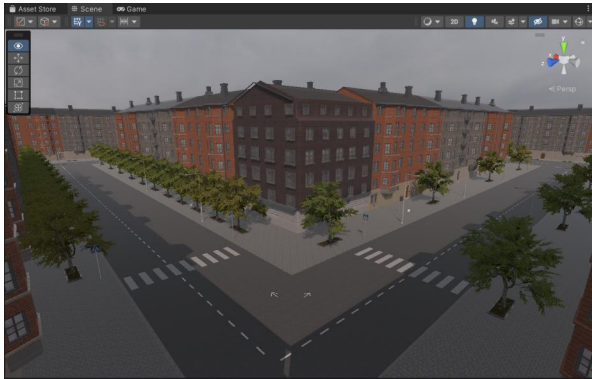


Figure B.2: HDG vs LDG

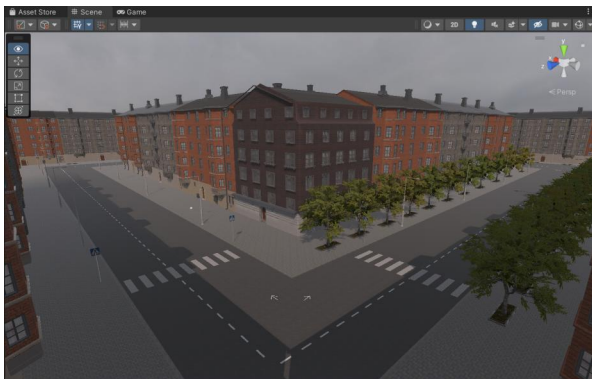


Figure B.3: HDG vs Empty



Figure B.4: HDG + SLA vs LDG + SLA

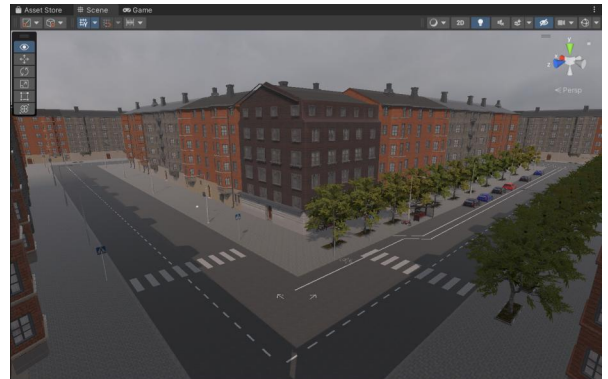


Figure B.5: HDG + SLA



Figure B.6: LDG vs HDG + SLA



Figure B.7: LDG vs LDG + SLA



Figure B.8: LDG vs SLA



Figure B.12: SLA vs HDG



Figure B.9: LDG vs Empty



Figure B.13: SLA vs Empty



Figure B.10: LDG + SLA vs SLA



Figure B.14: SLA + LDG vs HDG



Figure B.11: LDG + SLA vs Empty



Figure B.15: SLA+LDG vs Empty

C. PERCEIVED SAFETY QUESTIONNAIRE

Perceived safety

Please describe your experience regarding how safe you felt in the VR environment by choosing a number from the 7-point scale for each question. Read each question carefully and choose the number that best fits your experience. Use the whole scale. Answer the questions one by one, in order. There are no right or wrong answers - we are only interested in your personal experience.

How safe did you feel while walking on a street that had minimal greenery - 3 trees?

1234567

Not safe at all
☐
☐
☐
☐
☐
☐
☐
Very safe

Low density greenery - 3 trees

Figure C.1: Item Question LDG

How safe did you feel while walking on a street that had cars, bikes, a bike lane, a storefront, a bus stop etc?

1234567

Not safe at all
☐
☐
☐
☐
☐
☐
☐
Very safe

Street that had cars, bikes, a bike lane, a storefront, a bus stop etc

Figure C.3: Item Question SLA

How safe did you feel while walking on a street that had high density greenery - 8 trees?

1234567

Not safe at all
☐
☐
☐
☐
☐
☐
☐
Very safe

High density greenery - 8 trees

Figure C.2: Item Question HDG

How safe did you feel while walking on a street that had none of these things - it was only a window facade?

1234567

Not safe at all
☐
☐
☐
☐
☐
☐
☐
Very safe

Window facade

Figure C.4: Item Question Empty

D. 2AFC TEST

Table D.1: Contingency Tables

Handedness		Participant's Decision		Total
		L	R	
Left	Count	6.000	9.000	15.000
	Expected count	7.333	7.667	15.000
Right	Count	104.000	106.000	210.000
	Expected count	102.667	107.333	210.000
Total	Count	110.000	115.000	225.000
	Expected count	110.000	115.000	225.000

Table D.2: Chi-Squared Tests

	Value	df	p
X^2	0.508	1	0.476
N	225		

Table D.3: Log Odds Ratio

	Log Odds Ratio	95 % Confidence Intervals		p
		Lower	Upper	
Odds ratio	-0.386	-1.454	0.681	
Fisher's exact test	-0.385	-1.649	0.802	0.596